

Offline Semantic Guidance for Efficient Vision-Language-Action Policy Distillation

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Abstract

Billion-parameter Vision-Language-Action (VLA) policies have recently shown impressive performance in robotic manipulation, yet their size and inference cost remain major obstacles for real-time closed-loop control. We introduce **VLA-AD**, a distillation framework that uses a Vision-Language Model as an offline semantic supervisor to transfer large VLA teachers into lightweight student policies. Instead of relying only on low-level action imitation, VLA-AD augments teacher-provided 7-DoF action targets with high-level semantic guidance, including task phase anchors and multi-frame operating-direction descriptions. These auxiliary signals are used only during training: at test time, the student policy runs independently, with neither the VLA teacher nor the VLM required. We evaluate VLA-AD on three LIBERO benchmark suites. Using OpenVLA-7B as the teacher, our method produces a 158M-parameter student, yielding a $44\times$ reduction in model size while matching the teacher with only a 0.27% average relative gap. The resulting policy runs at 12.5 Hz on an RTX 4090, achieving a $3.28\times$ inference speedup over OpenVLA-7B. We further show that the same semantic distillation pipeline generalizes to a different $\pi_{0.5}$ -4B teacher, where the student outperforms the teacher on two suites and remains within 0.53% on `libero_goal`. Additional analysis indicates that phase-level supervision and multi-frame directional cues make the student less sensitive to noisy teacher actions, such as erroneous high-frequency gripper changes. Overall, VLA-AD demonstrates that offline semantic guidance from VLMs can substantially improve the efficiency, robustness, and deployability of VLA policy distillation.

1 Introduction

In recent years, robotic manipulation has undergone a paradigm shift with the emergence of large-scale vision-language-action (VLA) foundation models. By combining transformer architectures with large-scale vision-language pretraining and real-world robotic trajectories, these models have demonstrated strong multi-task generalization and zero-shot instruction-following capabilities [Brohan et al., 2023, Kim et al., 2024]. They represent an important step toward generalist robot policies that can execute

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diverse language-conditioned manipulation tasks in complex, unstructured environments. Despite these advances, deploying billion-parameter VLA models on real robotic platforms remains highly challenging. A major bottleneck is inference latency [Ahn et al., 2024]. Evaluating 7B-parameter or larger models on edge devices commonly used in robotics can require more than one second per control step [Brohan et al., 2023, Lin et al., 2026]. A common solution is to distill these large models into compact policies using standard behavioral cloning (BC) [Ross et al., 2011, Mandlekar et al., 2021]. However, this introduces a second challenge: vulnerability to compounding errors. Since single-task robotic datasets typically cover only a limited subset of possible states, cloned student policies can suffer from severe distribution shift during closed-loop execution [Chen et al., 2019]. As a result, a student may achieve high offline action-prediction accuracy but fail completely during online evaluation. Small prediction errors can accumulate over hundreds of timesteps, driving the robot into out-of-distribution (OOD) states that were not observed during training.

Recent advances suggest that vision-language models (VLMs) can provide a useful source of auxiliary supervision for addressing this problem. Although VLMs do not directly output low-level 7-DoF robot actions, they possess strong scene understanding, spatial reasoning, and commonsense physical priors [Huang et al., 2023, Bai et al., 2025]. Prior work has shown that aligning robotic policies with VLM-derived representations can improve semantic awareness and OOD generalization [Kachaev et al., 2025, Karamcheti et al., 2023]. This motivates the use of a VLM as an offline “knowledge translator”: it can extract task-relevant visual-semantic information from demonstrations and inject this information into a lightweight control policy, thereby bridging raw visual observations and high-level manipulation logic [Li et al., 2024, Ahn et al., 2022].

Motivated by this observation, we introduce **VLA-AD**, a VLM-assisted distillation framework for converting large VLA teachers into efficient closed-loop policies. VLA-AD uses a VLM as a parallel offline supervisor during training, while removing the VLM entirely at deployment. This allows the student to benefit from high-level semantic supervision without increasing inference latency. We evaluate VLA-AD through closed-loop experiments on the LIBERO benchmark and make the following contributions:

- **VLM-Assisted Zero-Overhead Distillation.** We propose VLA-AD, a parallel-supervision distillation framework that uses a VLM as an offline semantic supervisor for compressing billion-scale VLA teachers into compact student policies. The VLM provides phase and direction descriptions during training, but is removed entirely at deployment, introducing zero additional inference latency. With OpenVLA-7B as the teacher, VLA-AD allows a 158M-parameter student with $44\times$ parameter compression, matching the teacher within only 0.27% difference on average across three LIBERO suites. On an RTX 4090, the student reaches 12.5 Hz, giving a $3.28\times$ speedup over OpenVLA-7B.
- **Teacher-Agnostic Semantic Supervision Beyond Teacher Performance.** We show that VLA-AD is not tied to a specific teacher architecture: the same semantic-supervision pipeline transfers across both OpenVLA-7B and $\pi_{0.5}$ -4B teachers. Despite using a compact 158M-parameter student, VLA-AD matches OpenVLA-7B within only 0.27% on relative average across three LIBERO suites, and surpasses the $\pi_{0.5}$ -4B teacher on two of the three suites while remaining within 0.53% on the `libero_goal`. This indicates that VLM-generated phase and direction anchors capture transferable manipulation structure, allowing the student to learn closed-loop behaviours that are not merely bounded by the raw teacher action distribution.
- **Phase Anchors and Multi-Frame Direction for Robust Distillation.** We introduce a structured semantic-anchor mechanism that combines a 9-class phase descriptor with a multi-frame operating-direction signal. The phase descriptor provides a stable abstraction of the current manipulation stage, while the multi-frame direction signal resolves temporal ambiguity in visually similar operating states. This design gives the student consistent spatio-temporal context, helping it infer manipulation intent rather than overfitting to isolated frame-level observations.

Overall, VLA-AD provides a practical route toward deploying compact and robust VLA policies on real robotic platforms. By transferring semantic knowledge only during training, it improves closed-loop robustness while preserving the low-latency inference required for real-time control.

2 Related Work

2.1 Distillation of Large Robotic Policies

Deploying large-scale vision-language-action (VLA) foundation models [Kim et al., 2024, Brohan et al., 2023] on edge devices requires substantial model compression. Recent work has therefore explored VLA distillation as a way to reduce inference latency. TinyVLA [Wen et al., 2025] focuses on data-efficient architectural miniaturisation for high-frequency control, while CEED-VLA [Song et al., 2025] improves efficiency through early-exit decoding and consistency distillation. Other methods, such as VITA-VLA [Dong et al., 2025] and ActDistill [Ye et al., 2026], aim to transfer manipulation priors using action-expert distillation or dynamic action-guided routing.

Despite these advances, standard behavioural cloning (BC) remains vulnerable to compounding errors and covariate shift during closed-loop execution [Ross et al., 2011, Mandlekar et al., 2021]. Since single-task demonstration datasets usually cover only a limited range of states, small prediction errors can quickly drive the student policy into out-of-distribution (OOD) regions. Recent methods partially address this issue, but often rely on auxiliary action experts [Dong et al., 2025], dynamic routing mechanisms [Ye et al., 2026], or architectural changes that may complicate deployment [Song et al., 2025].

2.2 Phase Anchors and Temporal Context in Manipulation

Decomposing long-horizon manipulation tasks into discrete phases or semantic primitives is a well-established strategy for reducing control complexity [Ahn et al., 2022, Bharadhwaj et al., 2023]. Recent work has used large language models (LLMs) and vision-language models (VLMs) to generate high-level subgoals, task plans, or step-by-step instructions for guiding low-level robot policies [Huang et al., 2023]. In parallel, imitation learning methods have explored temporal context and action smoothing to address noisy demonstrations and short-term inconsistencies. Representative examples include observation-history conditioning, action chunking in ACT [Zhao et al., 2023], and denoising-based trajectory prediction in Diffusion Policy [Chi et al., 2024].

However, existing phase-based or hierarchical approaches often require multi-stage policy architectures or online queries to large models during execution, which conflicts with the low-latency requirements of edge deployment. Meanwhile, although action chunking and history-conditioned policies can reduce high-frequency action noise, they primarily rely on raw visual or proprioceptive histories and provide limited explicit semantic grounding. This makes lightweight students vulnerable to ambiguous or inconsistent teacher labels, such as spurious gripper oscillations observed in foundation VLA rollouts.

3 Method

3.1 Overall Architecture

We formulate VLA-AD as a parallel knowledge distillation framework designed to compress a computationally heavy VLA teacher (e.g., OpenVLA-7B or $\pi_{0.5}$ -4B) into a highly efficient student VLA. As illustrated in Figure 1, our offline training pipeline operates through a continuous, three-stage process. First, we collect expert manipulation trajectories by executing the teacher VLA across LIBERO tasks. We retain only the successful episodes to construct a curated dataset of raw visual observations paired with their corresponding 7-DoF expert actions, denoted as $a_{\text{teacher}} = (x, y, z, \theta, \phi, \psi, \text{gripper})$. Second, to bridge the representation gap between raw pixels and low-level continuous control, we leverage an off-the-shelf Vision-Language Model (VLM) to generate rich, phase-aware natural language annotations. This auxiliary semantic supervision is explicitly decoupled into two complementary streams: a single-frame prompt that directs the VLM to classify the current state into one of eight distinct non-operating phases (e.g., *grasping*, *transporting*), and a multi-frame prompt that queries the VLM across a 5-frame temporal window to extract the ongoing operating direction. Finally, the compact student VLA is trained end-to-end. It consumes the raw image observation alongside the VLM-generated single-frame and multi-frame text prompts as conditioned inputs, and is supervised directly by the teacher’s 7-DoF action targets. By establishing this parallel supervision mechanism, the student effectively grounds its continuous action predictions within robust, human-interpretable semantic anchors.

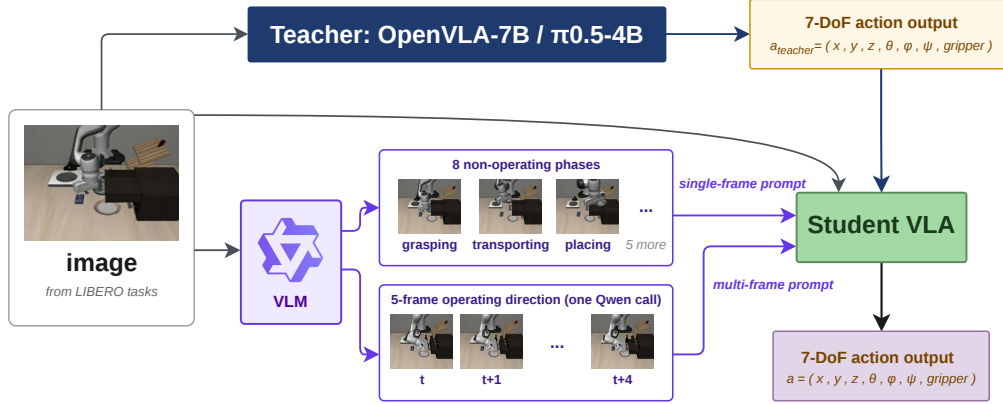


Figure 1: VLA-AD: Overall VLM-Assisted Distillation pipeline. A lightweight Student VLA is trained via parallel offline supervision from a frozen teacher VLA (e.g., OpenVLA-7B or $\pi_{0.5-4B}$) and a Vision-Language Model (VLM).

3.2 Motivation for Dual-Path Supervision

At each timestep, the student is required to output a 7-DoF action vector (comprising 3D translation increments $(\Delta x, \Delta y, \Delta z)$, 3D rotation increments $(\Delta r_x, \Delta r_y, \Delta r_z)$, and a 1D gripper state) while simultaneously maintaining a high-level understanding of the ongoing task phase. However, neither of our off-the-shelf supervisory sources provides a complete picture independently. On one hand, the teacher VLA (e.g., OpenVLA-7B or $\pi_{0.5-4B}$) supplies precise, per-frame 7-DoF action values, dictating exactly how much to move and whether to actuate the gripper. Yet, this continuous supervision lacks semantic grounding; the student receives numerical targets but remains unaware of the overarching task phase or the logical rationale behind discrete events, such as a sudden gripper closure. On the other hand, the VLM (Qwen2.5-VL) provides rich natural language descriptions that characterize the gripper state, relative object positions, and the current phase. However, the VLM operates purely in the semantic domain and cannot predict the low-level continuous action signals required for control.

In essence, the teacher demonstrates how to execute the manipulation, while the VLM explains what the current action is achieving. The student must synthesize both modalities during training so that it can independently generate temporally coherent and numerically accurate 7-DoF actions during evaluation (when both the teacher and VLM are removed). To achieve this, we introduce a dual-path training scheme. Both paths are supervised directly by the teacher’s action targets. Formally, let d_t denote the VLM-generated description for frame t , and $a_t^* \in \{0, \dots, 255\}^7$ the per-dimension discretized teacher action. The total dual-path loss is defined as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{full}}(x_t, \tau, d_t) + \alpha \cdot \mathcal{L}_{\text{img}}(x_t, \tau), \quad (1)$$

where each per-path loss is a dimension- and chunk-weighted cross-entropy defined over a prediction horizon of K steps:

$$\mathcal{L}_{\bullet} = \sum_{k=1}^K \sum_{j=1}^7 w_j \cdot \text{CE}(\pi_S^{(k,j)}(\cdot | \bullet), a_{t+k,j}^*). \quad (2)$$

Here, the weights $w = (1, 1, 1, 2, 2, 2, 1)$ are introduced to compensate for the smaller magnitude of the rotation channel. The full path, $\mathcal{L}_{\text{full}}$, utilizes the visual observation (x_t) , task instruction (τ) , and semantic description (d_t) . Conversely, the image-only path, $\mathcal{L}_{\text{img}}$, masks the description channel. This masking forces the visual representation to remain self-sufficient and prevents the student from collapsing into a description-only shortcut. The hyperparameter α dictates the weight of this auxiliary path.

3.3 Phase-Anchored Visual Description

Allowing a VLM like Qwen2.5-VL to freely describe the visual scene (as explored in our early pilot studies) introduces significant linguistic stochasticity. For the exact same manipulation stage, the VLM might output vastly different phrasing, such as “the robot is grabbing” versus “reaching for.” This descriptive noise is severely amplified when distilling into a compact 158M-parameter student. Unlike the 7B-parameter teacher, the lightweight student lacks the redundant capacity to absorb such linguistic variance. Consequently, representation-level inconsistencies directly corrupt the language conditioning, degrading the overall distillation quality.

To suppress the inherent randomness of the VLM, we introduce **phase-Anchored Visual Description**. Specifically, we employ a heuristic rule-based classifier (utilizing signals such as gripper state, 3D velocity, and task progression) to assign a fixed phase label to each frame. This phase label, along with its semantic definition, is then injected into the VLM’s prompt. This constraint forces Qwen to explicitly use the assigned phase vocabulary while describing the relative object positions and the ongoing action. As a result, the description for every frame is anchored to a consistent semantic coordinate, leaving the VLM only the degree of freedom to clearly articulate the scene context in two to four natural language sentences.

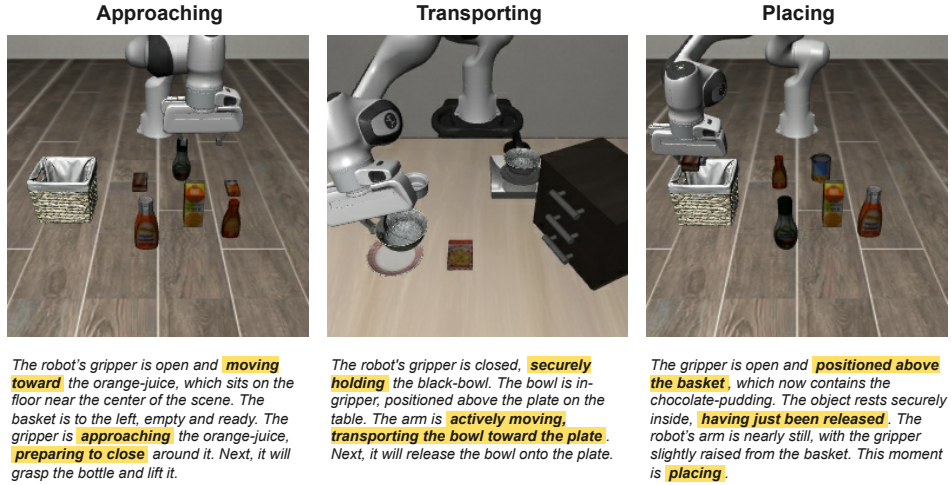


Figure 2: Qualitative examples of Qwen2.5-VL phase-anchored descriptions on three OpenVLA-7B rollouts from `libero_object`. Each panel shows a single frame together with the description Qwen produces under phase-anchored prompt for the phases *approaching*, *transporting*, and *placing*. Highlighted text marks the action-semantic vocabulary that is tied to the phase label.

During this phase-anchored annotation process, we inadvertently identified several anomalous behaviors exhibited by the teacher policy, likely stemming from noise in its original training distribution. As prominently illustrated in Figure 4, the OpenVLA-7B teacher frequently exhibits spurious, high-frequency gripper oscillations, emitting contradictory open/close signals even when the visual scene remains nearly identical. A standard behavioral cloning student would blindly overfit to these self-revoking labels. However, our phase classifier categorizes these anomalous sequences into stable, continuous phases (e.g., *holding*). By combining these stable phase anchors with the semantic VLM descriptions, the student learns to prioritize the broader manipulation context over the noisy, frame-level action targets. Consequently, rather than inheriting the teacher’s instability, the lightweight student becomes robust enough to smooth out these expert errors, outputting a significantly more consistent and logical control policy.

The number of distinct phases is a critical hyperparameter. We evaluated six different taxonomy granularities (3, 5, 7, 9, 11, and 13 phases) by measuring the Coefficient of Variation (CV) to assess how evenly the phases were distributed across the six LIBERO suites. The 9-phase taxonomy achieved

the lowest deviation ($|CV - 1| = 0.14$), most closely approximating a Gaussian-like distribution. It strikes an optimal balance: it is neither too coarse to conflate distinct manipulation stages, nor too fine to result in long-tailed, undersampled phases. Consequently, we adopted a 9-phase taxonomy: *idle*, *approaching*, *grasping*, *transporting*, *holding*, *placing*, *operating*, *regrasping*, and *completed*.

It is important to note that the specific lexical choices (e.g., “grasping” versus “transporting”) are secondary. The student is not merely learning these isolated vocabulary words; rather, it is learning the comprehensive semantic representation that accompanies each phase label. This includes the VLM’s full description of the gripper’s current action, the relative spatial coordinates of the target object, spatial constraints, and the anticipated next step. The phase labels serve primarily as a linguistic anchor to standardize these rich, context-aware descriptions.

3.4 Multi-Frame Operating Direction

While single-frame Qwen annotations suffice for the majority of manipulation phases, the *operating* phase (e.g., opening a drawer or turning a knob) presents a unique challenge: directional ambiguity. From a single static frame, the direction of motion is often indiscernible. For example, a partially open drawer appears identical whether it is actively being pulled open or pushed closed; the spatial coordinates of the drawer, the robot’s proprioceptive posture, and the gripper state are visually indistinguishable. This ambiguity introduces confounding noise into the student’s learned policy. Because the supervision signal for the gripper channel must also implicitly capture the precise timing of an object’s release, a lack of clear directional context can severely perturb the student’s action predictions.

To resolve this, we introduce a multi-frame operating direction signal, formulated as a tuple: (element, direction). The element specifies the targeted object or part being manipulated (e.g., *drawer-handle*, *knob*, *bowl*), while the direction characterizes its dynamic motion (e.g., *outward*, *clockwise*, *downward*).

While the other eight phases continue to rely exclusively on single-frame annotations, the *operating* phase triggers a broader temporal context window. When a continuous sequence of frames is classified as *operating*, we extract the entire segment and uniformly sample five keyframes spanning its beginning, middle, and end. These five frames are fed jointly to the VLM. By observing the visual progression across this temporal span, Qwen accurately infers the underlying (element, direction) tuple. This resulting tuple is then broadcast to every frame within that specific operating segment. For these frames, the final text description fed to the student is constructed by concatenating the phase label with the multi-frame tuple, providing a temporally stable and disambiguated semantic anchor.

4 Experiments

4.1 Experimental Setup

We evaluate VLA-AD on three LIBERO suites: `libero_object`, `libero_spatial`, and `libero_goal` [Liu et al., 2023]. Following the standard OpenVLA protocol, we conduct 20 closed-loop episodes across 10 tasks per suite, yielding 600 total evaluation trials with a maximum horizon of 520 steps. To demonstrate that our distillation pipeline is teacher-agnostic, we use two expert models, OpenVLA-7B and $\pi_{0.5}$ -4B, to supervise a unified 158M-parameter student policy. The student architecture comprises a Long-CLIP encoder [Zhang et al., 2024], LoRA adapters [Hu et al., 2021], and a tri-stream MLP head that predicts an action chunk of length $K = 5$ [Zhang et al., 2025]. This achieves $44\times$ and $25\times$ parameter compression relative to the respective teachers. To ensure a fair comparison, we re-evaluate both teachers under our exact closed-loop protocol rather than relying on officially reported metrics. For each teacher, we compare a *no-VLM* baseline, which is conditioned solely on raw images and instructions, against our proposed *with-VLM* method, which integrates phase descriptions and multi-frame directions. For the latter, we sweep the dual-path hyperparameter $\alpha \in \{0.3, 0.5, 0.8, 1.0\}$. The student is trained using AdamW for 30 epochs with a batch size of 32. By injecting rank-8 LoRA modules into the Long-CLIP transformer blocks and freezing the remaining backbone, we reduce the number of trainable parameters to 8.6M. This requires approximately 22 GPU-hours per student, representing a $10\times$ – $20\times$ reduction in computational cost compared with full OpenVLA fine-tuning [Kim et al., 2024]. Finally, offline annotation using the Qwen2.5-VL API is highly economical. Annotating an entire suite of 81,000 frames costs approximately 7 USD,

representing a negligible overhead and remaining more than two orders of magnitude cheaper than fine-tuning a 7B-parameter teacher on the same data.

4.2 Cross-Teacher Generalization

We evaluate whether VLA-AD generalizes across different VLA teachers using the two aggregate studies in Tables 1 and 2. With OpenVLA-7B as the teacher, the 158M-parameter student matches teacher-level performance across all three LIBERO suites (Table 1), under suite-specific best α values from a sweep over $\alpha \in \{0.3, 0.5, 0.8, 1.0\}$. Compared with the no-Qwen baseline, VLM supervision brings the largest gain on `libero_goal`, suggesting that phase and direction descriptions are most useful for multi-stage tasks where the student must track sequential subgoals such as approaching, grasping, operating, and placing. To test teacher-agnostic transfer, we replace OpenVLA-7B with $\pi_{0.5}$ -4B while reusing the same suite-specific α values selected from the OpenVLA sweep (Table 2). The with-Qwen student outperforms the $\pi_{0.5}$ teacher on `libero_object` and `libero_spatial` while staying close on `libero_goal`. The no-Qwen baseline is already strong on object and spatial tasks, indicating that supervised distillation with Long-CLIP features and short-horizon action chunking ($K = 5$) can produce a robust closed-loop policy. In contrast, the goal suite benefits more clearly from VLM descriptions, where Qwen supervision nearly recovers the teacher’s level. Overall, these results show that VLA-AD is not tied to a specific teacher. The same α choices transfer from OpenVLA-7B to $\pi_{0.5}$, suggesting that the loss balance is more strongly determined by the task suite than by the teacher architecture. In particular, object and goal tasks remain effective with $\alpha = 1.0$, where visual control should dominate and VLM descriptions act as semantic anchors. Spatial tasks prefer $\alpha = 0.5$, reflecting the greater value of explicit relational language for resolving spatial configurations.

To test teacher-agnostic transfer, we replace OpenVLA-7B with $\pi_{0.5}$ -4B while reusing the same suite-specific α values selected from the OpenVLA sweep. As shown in Table 2, the with-Qwen student reaches 96.5%, 93.0%, and 94.0% on the three suites, outperforming the $\pi_{0.5}$ teacher on `libero_object` and `libero_spatial` by +7.22 and +6.90 percentage points, while remaining within 0.5 points on `libero_goal`. The no-Qwen baseline is already strong on object and spatial tasks, indicating that supervised distillation with Long-CLIP features and short-horizon action chunking ($K = 5$) can produce a robust closed-loop policy. In contrast, the goal suite benefits more clearly from VLM descriptions: Qwen supervision recovers the drop from 90.0% to 94.0%, nearly matching the teacher.

Overall, these results show that VLA-AD is not tied to a specific teacher. The same α choices transfer from OpenVLA-7B to $\pi_{0.5}$, suggesting that the loss balance is more strongly determined by the task suite than by the teacher architecture. In particular, object and goal tasks remain effective with $\alpha = 1.0$, where visual control should dominate and VLM descriptions act as semantic anchors. Spatial tasks prefer $\alpha = 0.5$, reflecting the greater value of explicit relational language for resolving spatial configurations.

4.3 The Granularity Dilemma and Phase Selection

The phase descriptor acts as a semantic anchor that tells the student which stage of the manipulation task it is executing. Its effectiveness, however, depends on choosing an appropriate taxonomy granularity. If the taxonomy is too coarse, multiple behaviors collapse into a dominant class, making the phase signal uninformative. If it is too fine, many categories become sparse or empty, leaving the student with poorly supported labels. We therefore seek a phase distribution that is neither uniform nor single-peaked, but instead matches the natural structure of manipulation trajectories: long movement stages such as approaching and transporting occupy many frames, while short transition stages such as grasping, releasing, and placing occur less frequently.

Table 1: OpenVLA-7B closed-loop success rate (% , 200 trials/cell) on 3 LIBERO in domain suites.

Method	object	spatial	goal	Mean
OpenVLA-7B	62.0	79.0	81.0	74.0
VLA-AD (no-Qwen)	62.0	77.0	61.0	66.7
VLA-AD (with-Qwen)	62.5	79.5	79.5	73.8
VLA-AD vs. Teacher	+0.81%	+0.63%	-1.85%	-0.27%

Table 2: $\pi_{0.5}$ closed-loop success rate (% , 200 trials/cell) on 3 LIBERO in domain suites.

Method	object	spatial	goal	Mean
$\pi_{0.5}$ (teacher)	90.0	87.0	94.5	90.5
VLA-AD (no-Qwen)	94.5	94.0	90.0	92.8
VLA-AD (with-Qwen)	96.5	93.0	94.0	94.5
vs. Teacher	+7.22%	+6.90%	-0.53%	+4.42%

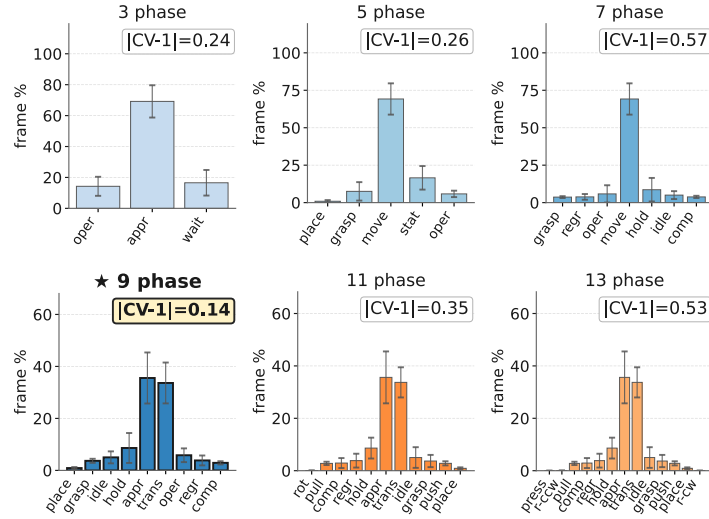


Figure 3: We compare phase vocabularies from 3 to 13 categories, averaged over six teacher–suite rollouts. Each panel shows the mean per-phase frame share with standard-deviation error bars across rollouts, reordered so that the largest bin is centered. Coarser taxonomies merge major motion stages into a single dominant bin, whereas finer taxonomies introduce sparse or nearly empty operating sub-categories.

To quantify this balance, we use a modified coefficient of variation, CV , computed from the cross-category frame frequencies, and report $|CV - 1|$ as the deviation from a target peaked distribution. Lower values indicate a better balance between dominant but meaningful phases and sufficiently populated tail categories. As shown in Figure 3, the 9-phase taxonomy provides the best granularity. Coarser taxonomies with 3, 5, and 7 phases merge major motion stages, producing a dominant bin of roughly 69% of all frames and yielding larger deviations ($|CV - 1| = 0.24, 0.26$, and 0.57). In contrast, the 9-phase taxonomy separates this behavior into two balanced phases, approaching (35.6%) and transporting (33.6%), while keeping the remaining phases non-trivial. It achieves the lowest deviation, $|CV - 1| = 0.14$. Finer taxonomies with 11 and 13 phases over-subdivide operating behaviors into categories such as push, pull, rotate, and press, many of which are rarely populated. This creates long-tailed distributions with higher deviations ($|CV - 1| = 0.35$ and 0.53). The narrow error bars in Figure 3 further show that this 9-phase sweet spot is consistent across the six teacher–suite rollouts, suggesting that it reflects a stable property of the manipulation data rather than a teacher-specific artifact.

4.4 Inference Efficiency

The 158M-parameter student significantly outperforms both teachers in inference speed and memory footprint, as shown in Table 3. On an NVIDIA RTX 4090, the OpenVLA-distilled student achieves 12.5 Hz, corresponding to a $3.28\times$ speedup over the 7B teacher, which runs at 3.8 Hz, while using $44\times$ fewer parameters. Similarly, the $\pi_{0.5}$ -distilled student reaches 13.2 Hz, providing a $25\times$ parameter compression compared with its 4B teacher. By delivering more than $3\times$ the control frequency of the OpenVLA baseline, our student model enables high-frequency, real-time closed-loop control on consumer-grade hardware.

Table 3: End-to-end inference time on RTX 4090.

Model	Params	sec/step	Hz	Speedup
OpenVLA-7B	7B	0.262	3.8	$1\times$
VLA-AD(OpenVLA)	158M	0.080	12.5	$3.28\times$
$\pi_{0.5}$ -4B	4B	0.172	5.81	$1.52\times$
VLA-AD($\pi_{0.5}$)	158M	0.076	13.2	$3.45\times$

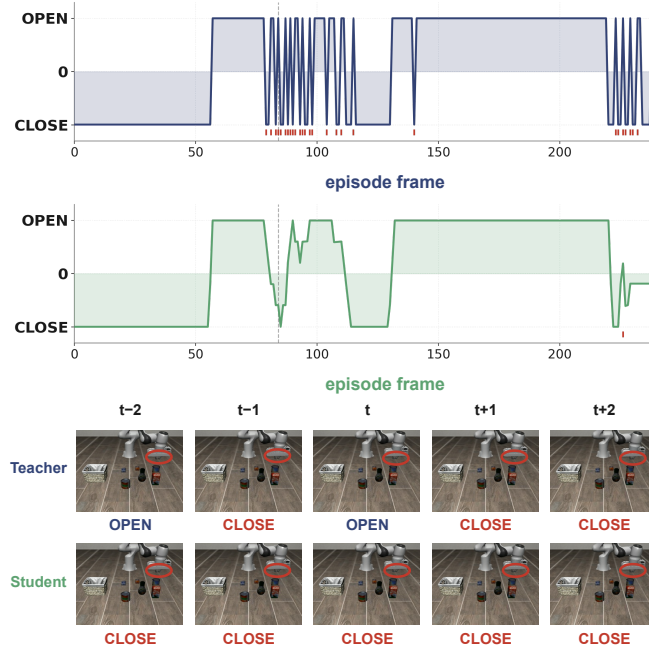


Figure 4: Spurious gripper-command oscillation in OpenVLA-7B teacher rollouts on `libero_object` task 0. **Top:** Per-frame gripper command for episode 15; red markers denote sign reversals that revert within ≤ 2 frames. The blue line shows the teacher’s noisy output, while the green line shows the student’s smoothed and robust predictions. **Bottom:** Five consecutive frames around the flip marked by the dashed line.

4.5 Robust Distillation from Noisy Teachers

As illustrated in Fig. 4, the OpenVLA-7B teacher exhibits approximately 3% systematic noise in its gripper signals, characterized by high-frequency spurious flips, such as 27 reversals within 240 frames. Such noise can prevent standard behavior-cloning students from converging to a stable policy, leading instead to oscillation or policy collapse. VLA-AD mitigates this issue through VLM-generated phase descriptions, which provide a stable semantic anchor, such as “grasping”, across continuous segments of more than 50 frames.

While the teacher’s per-frame commands oscillate, the invariant phase label encourages the student to prioritize the underlying physical process rather than single-frame noise. Audit results show that the student reduces spurious flips by $9\times$ compared with the teacher, decreasing from 27 flips to 1. The qualitative comparison in Fig. 4 (bottom) further demonstrates that the student maintains a smooth and consistent gripper strategy across nearly identical visual frames, showing that phase anchors significantly improve distillation robustness against teacher inaccuracies. The visual scene is nearly identical, yet the teacher emits contradictory gripper signals (OPEN/CLOSE/OPEN). In contrast, the student maintains a consistent action. Across the full task, we observe 242 spurious flips in 8,223 frames, meaning that approximately 3% of supervisory targets are self-revoking.

5 Conclusion

This work examines how offline VLM supervision can improve the distillation of large VLA teachers into compact closed-loop robot policies. Instead of relying solely on frame-level action imitation, VLA-AD augments teacher actions with phase anchors and multi-frame direction cues, providing the student with a stable semantic description of the ongoing manipulation process. Because the VLM is used only during training, this additional supervision does not affect deployment latency.

The experimental results show that such semantic structure can substantially improve the reliability of lightweight students. VLA-AD closely matches OpenVLA-7B across three LIBERO suites with a 158M-parameter model and transfers to a distinct $\pi_{0.5}$ -4B teacher, suggesting that the learned semantic anchors capture manipulation structure beyond a single teacher’s action distribution. The gripper-noise analysis further indicates that phase-level supervision can regularize imperfect teacher demonstrations and help the student produce smoother closed-loop behavior. These findings suggest a broader role for VLMs in robotics: rather than serving only as online planners or direct controllers, they can act as scalable offline annotators that inject semantic structure into policy learning. Future work should evaluate this paradigm on real-robot deployments, richer multi-task datasets, and adaptive annotation strategies that can better capture uncertainty and task-specific temporal structure.

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Appendix

Supplementary video: per-frame gripper-command comparison

We provide `gripper_comparison_video.mp4` (30 s, 8 fps, 240 frames) as a side-by-side per-frame visualization of the gripper command emitted by the OpenVLA-7B teacher (left panel, navy) and by our 158M Long-CLIP+LoRA student distilled from its rollouts (right panel, sage), played frame-by-frame through an episode of `libero_object` task 0. The teacher’s command is its actually-executed closed-loop action; the student is evaluated open-loop on the same 240 RGB frames. Each panel reports the discrete OPEN/CLOSE label, continuous gripper value, and a running count of spurious sign-reverts. Final tally: teacher 27, student 1.

Phase taxonomy granularity ablation

The number of phases in our taxonomy is a critical hyperparameter (Section 4.3). To complement the $|CV - 1|$ analysis in Fig. 3, we report next-action token validation accuracy for the same 158M Long-CLIP+LoRA student trained under each of the six taxonomies (3, 5, 7, 9, 11, 13 phases), holding $\alpha = 1.0$ and 30 epochs fixed. Coarser taxonomies (≤ 7) under-specify action semantics; finer ones (≥ 11) fragment otherwise unitary actions. The 9-phase taxonomy maximizes mean accuracy on all three suites.

Table 4: 158M Long-CLIP+LoRA student per-suite validation accuracy (%) across phase taxonomy granularity, $\alpha = 1.0$. Each cell = next-action token accuracy on a held-out 10% split of training rollouts. Bold = highest-mean row.

# phases	object	spatial	goal	mean
3	54.2	52.0	53.5	53.2
5	58.5	56.0	57.5	57.3
7	61.0	58.5	60.0	59.8
9	64.8	62.5	63.5	63.6
11	60.5	58.0	59.5	59.3
13	56.0	54.0	55.0	55.0

Per-task results: OpenVLA-7B teacher with α sweep

Table 5 expands the aggregate OpenVLA-7B results in Table 1 into per-task closed-loop success rates across the no-Qwen baseline and the four α values swept (0.3, 0.5, 0.8, 1.0). Each cell is the success rate over 20 trials of one (suite, task) configuration; bold marks the α -cells where the with-Qwen student strictly outperforms its same-row no-Qwen baseline. The breakdown surfaces task-level variance hidden by the aggregate mean: gains concentrate on tasks where the phase descriptor disambiguates the manipulation target (e.g., goal T7–T9), while a small number of object tasks regress under strong VLM weighting.

Table 5: OpenVLA-7B 30ep per-task closed-loop success rate (%). Each cell = 20 trials. Bold = α -cell strictly greater than same-row no-Qwen.

Suite	Task	no-Qwen	$\alpha = 0.3$	$\alpha = 0.5$	$\alpha = 0.8$	$\alpha = 1.0$
object	T0	40.0	10.0	20.0	15.0	20.0
	T1	50.0	50.0	60.0	70.0	50.0
	T2	65.0	50.0	65.0	60.0	70.0
	T3	45.0	50.0	55.0	50.0	55.0
	T4	95.0	95.0	95.0	95.0	85.0
	T5	40.0	30.0	35.0	25.0	55.0
	T6	75.0	65.0	70.0	65.0	70.0
	T7	65.0	70.0	55.0	65.0	65.0
	T8	75.0	75.0	45.0	50.0	70.0
	T9	70.0	75.0	55.0	65.0	85.0
spatial	T0	85.0	80.0	90.0	90.0	95.0
	T1	70.0	85.0	95.0	85.0	80.0
	T2	75.0	65.0	75.0	80.0	65.0
	T3	90.0	85.0	95.0	80.0	95.0
	T4	50.0	45.0	45.0	55.0	35.0
	T5	100.0	80.0	90.0	90.0	80.0
	T6	100.0	95.0	100.0	100.0	100.0
	T7	90.0	95.0	85.0	70.0	75.0
	T8	75.0	80.0	75.0	80.0	75.0
	T9	35.0	25.0	45.0	40.0	45.0
goal	T0	60.0	60.0	70.0	55.0	70.0
	T1	90.0	80.0	85.0	75.0	90.0
	T2	70.0	95.0	85.0	85.0	90.0
	T3	25.0	50.0	30.0	25.0	40.0
	T4	70.0	90.0	85.0	85.0	95.0
	T5	80.0	90.0	90.0	70.0	90.0
	T6	60.0	70.0	70.0	70.0	90.0
	T7	80.0	95.0	100.0	85.0	90.0
	T8	35.0	70.0	70.0	75.0	75.0
	T9	40.0	60.0	55.0	65.0	65.0

Per-task results: $\pi_{0.5}$ -4B teacher at suite-best α

Table 6 reports per-task closed-loop success rates of the $\pi_{0.5}$ -4B-distilled student (cf. Table 2 for aggregates), comparing the no-Qwen baseline against the with-Qwen variant at the suite-specific best α selected from the OpenVLA sweep (Table 1): $\alpha = 1.0$ for object and goal, $\alpha = 0.5$ for spatial. Each cell is 20 trials; bold marks tasks where the with-Qwen student strictly outperforms its no-Qwen counterpart. Improvements concentrate on tasks that involve sustained grasping or precise placement, consistent with the design intent of the phase-anchored description.

Table 6: $\pi_{0.5}$ 30ep per-task closed-loop success rate (%). Each cell = 20 trials. Bold = with-Qwen strictly greater than no-Qwen.

Suite	Task	no-Qwen	with-Qwen
object ($\alpha = 1.0$)	T0	100.0	100.0
	T1	95.0	100.0
	T2	100.0	85.0
	T3	90.0	95.0
	T4	100.0	100.0
	T5	80.0	95.0
	T6	90.0	100.0
	T7	95.0	95.0
	T8	100.0	100.0
	T9	95.0	95.0
spatial ($\alpha = 0.5$)	T0	95.0	95.0
	T1	100.0	95.0
	T2	100.0	95.0
	T3	90.0	95.0
	T4	100.0	100.0
	T5	75.0	70.0
	T6	100.0	100.0
	T7	95.0	100.0
	T8	90.0	85.0
	T9	95.0	95.0
goal ($\alpha = 1.0$)	T0	85.0	95.0
	T1	95.0	100.0
	T2	85.0	95.0
	T3	80.0	80.0
	T4	90.0	100.0
	T5	95.0	90.0
	T6	80.0	85.0
	T7	100.0	100.0
	T8	95.0	100.0
	T9	95.0	95.0

Limitations. Several limitations should be considered when interpreting our results. (i) All closed-loop evaluations use a single random seed; statistical reliability is supported by 200 trials per (suite, condition) cell rather than by multi-seed variance. (ii) We evaluate only on three in-domain LIBERO suites; out-of-distribution variants (e.g., LIBERO-PRO), real-robot deployment, and long-horizon cross-suite settings remain untested. (iii) The teacher-agnostic claim rests on $n=2$ teachers (OpenVLA-7B and $\pi_{0.5}$ -4B); generalization to weaker, larger, or architecturally distinct VLA teachers is unverified. (iv) The dual-path weight α is selected per suite (object/goal = 1.0, spatial = 0.5), so a small sweep is still required when applying VLA-AD to a new task suite. (v) Our distillation set retains only successful teacher episodes, so the student never observes teacher failure modes, potentially limiting recovery from out-of-distribution states. (vi) Inference latency is benchmarked on an RTX 4090 desktop GPU; performance on resource-constrained edge devices (e.g., Jetson AGX Orin) has not been measured. (vii) The rule-based phase classifier relies on LIBERO-specific gripper and proprioceptive signals and would require redesign for environments without comparable state access.

Broader Impacts. VLA-AD aims to make large vision-language-action policies more efficient at deployment, which has both positive and negative societal implications. On the positive side, reducing parameter count by an order of magnitude and roughly tripling inference throughput lowers the energy, hardware, and cost barriers to deploying robotic assistants in low-resource environments such as small clinics, warehouses, and educational labs, and reduces the carbon footprint of ongoing robotic operation. On the negative side, more efficient and easily deployable manipulation policies could accelerate the displacement of human labor in repetitive physical tasks, and any safety failure in a distilled student — including failure modes silently inherited from the teacher — could cause physical harm if deployed without adequate supervision. We have not evaluated VLA-AD on real robots, and we recommend that any real-world deployment include human-in-the-loop verification,

formal safety envelopes, and continued monitoring for inherited teacher errors such as the spurious gripper oscillations analyzed.